

16 Work done needed is the total electrical potential energy at the centre of the square.

$$\begin{aligned} &= \frac{Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{2}}\right)} + \frac{-2Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{2}}\right)} + \frac{3Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{2}}\right)} + \frac{-4Q}{4\pi\epsilon_0 \left(\frac{a}{\sqrt{2}}\right)} \\ &= \frac{-\sqrt{2}Q}{2\pi\epsilon_0 a} \end{aligned}$$

Answer: C

17 Electric field points in the direction of decreasing potential. Electric potential energy is